

Fix Click Detection

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<https://chatgpt.com/share/6ac679b8-9158-83ee-bcc6-78f04ee8b652>

USER

Balloon Pop Lab

This project is a single-topic Balloon Pop game using **Pygame**. It introduces students to size-aware click detection, entity variety, a lives/miss system, and round timing, using a small, readable object-oriented codebase.

What's Provided

A working Balloon Pop game with:

- Balloons that spawn at the top at random sizes and fall toward the bottom at random speeds
- Clicking a balloon pops it and awards points; letting one fall past the bottom is currently ignored
- A running score display

It has **one deliberate bug** and **three features** left for you to build. You are expected to **analyze, interact with an AI assistant**, and **complete/fix** the game to make it fully functional and more interesting.

Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this lab.

Getting Started

Setup

1. Make sure you have Python 3.10+ installed.
2. Install dependencies:

```
pip install -r requirements.txt
```

3. Run the game:

```
python main.py
```

Controls: Left-click a balloon to pop it before it reaches the bottom.

Tasks to Complete

Each task must be completed using an iterative process involving LLM suggestions and your critical code review.

Task 1: Fix the click-detection bug

A click is supposed to pop a balloon whenever it lands reasonably close to that balloon's center - anywhere within its visible circle. In the current build, `check_pop` (in `game/click_detection.py`) computes the *squared* distance from the click to the balloon's center, but compares it directly against the balloon's plain (non-squared) `radius`. Squared distance grows much faster than linear distance, so this comparison only succeeds when the click lands within a few pixels of dead-center - anywhere close to the visible edge of the balloon fails to register at all, even though it clearly looks like a hit. Fix the check so it compares actual distance to the balloon's actual radius.

Task 2: Implement different balloon types

Introduce at least three balloon types: a normal balloon (regular points), a bonus balloon (more points), and a penalty balloon (which reduces the score, or has some other clearly defined negative effect). Give each type its own color so they're easy to tell apart, and make sure the correct effect happens when each type is popped.

Task 3: Implement a lives/miss system

Add 3 lives. Whenever a balloon reaches the bottom of the screen without being popped, the player should lose a life. Display the remaining lives, and end the game once they reach zero. Popping a balloon should never cost a life, no matter its type.

Task 4: Implement a timed round

Add a 30-second countdown for the round. Display the remaining time on screen. Once it reaches zero (or lives run out, once Task 3 is done), stop spawning and accepting balloons, show the final score clearly, and provide a way to start a new round with the score, lives, and timer all reset.

Expected Behavior

- Clicking on a balloon should reliably pop it - not just when you happen to click exactly on its center pixel. Test this by clicking normally, the way you would in actual play, not by aiming with pixel-perfect precision.
- Balloon types are visually distinguishable and correctly affect the score when popped.
- Letting a balloon fall past the bottom costs a life; popping one never does, regardless of type.
- The round ends when time runs out or lives reach zero, whichever comes first, with the final score shown clearly and a way to start again.

Folder Structure

```
balloon-pop/  
%%%   main.py  
%%%   requirements.txt  
%%%   game/  
%       %%%   game_engine.py  
%       %%%   balloon.py  
%       %%%   click_detection.py  
%       %%%   renderer.py  
%%%   README.md
```

Submission Checklist

Submission is only the following three things:

- A 10-second video of gameplay **before** your changes, showing the bug/broken behavior
- A 10-second video of gameplay **after** your changes, showing the bug fixed and the new features working
- The Chat/LLM used page link, with the complete chat history
1. fork into your account 2. get clone \<link> 3. video of before and after 4. prompt llm to make changes (after teach task we have to commit we need to have 4 commits) 5. to commit and push to github

ASSISTANT

Here is the **exact workflow** you should follow for your PESU Lab 4 submission.

The lab requires:

- Fork repo
- Clone to your laptop
- Record **before video**
- Fix bug + add features
- Make **separate commits for each task**
- Record **after video**
- Push to GitHub
- Upload chat history PDF

Step 1: Fork the Repository

Open:

Click:

Fork !' Create Fork

This creates:

TEXT

```
https://github.com/YOUR_USERNAME/05_balloon_pop
```

Example:

TEXT

```
https://github.com/MadhumithaDhondi/05_balloon_pop
```

Step 2: Clone Repository

Open terminal:

BASH

```
cd Desktop
```

Clone:

BASH

```
git clone https://github.com/YOUR_USERNAME/05_balloon_pop.git
```

Example:

BASH

```
git clone https://github.com/MadhumithaDhondi/05_balloon_pop.git
```

Go inside:

BASH

```
cd 05_balloon_pop
```

Step 3: Install and Run

Install pygame:

BASH

```
pip install -r requirements.txt
```

Run:

BASH

```
python main.py
```

or:

BASH

```
python3 main.py
```

Step 4: Record BEFORE Video (10 sec)

Show:

- Balloon spawning
- Clicking near edge not working
- No lives
- No timer
- One balloon type only

Save:

TEXT

```
before.mp4
```

Step 5: Task 1 — Fix Click Bug

Current bug:

PYTHON

```
if distance_squared <= balloon.radius:
```

Wrong because:

TEXT

distance² compared with radius

Correct:

PYTHON

```
if distance_squared <= balloon.radius * balloon.radius:
```

OR:

PYTHON

```
distance = (dx*dx + dy*dy) ** 0.5
```

```
if distance <= balloon.radius:
```

Bug exists here:

Commit 1

BASH

```
git add .  
git commit -m "Task 1: Fixed balloon click detection bug"  
git push
```

Step 6: Task 2 — Balloon Types

Add:

Normal Balloon

PYTHON

```
Color: Pink  
Points: +10
```

Bonus Balloon

PYTHON

```
Color: Green  
Points: +25
```

Penalty Balloon

PYTHON

Color: Red
Points: -10

Starter code currently has only one type:

Commit 2

BASH

```
git add .  
git commit -m "Task 2: Added normal bonus and penalty balloons"  
git push
```

Step 7: Task 3 — Lives System

Add:

PYTHON

```
self.lives = 3
```

When balloon misses:

PYTHON

```
lives -= 1
```

Display:

TEXT

```
Lives: 3
```

Game over:

PYTHON

```
if lives <= 0:  
    game_over = True
```

Commit 3

BASH

```
git add .  
git commit -m "Task 3: Added lives system and game over"  
git push
```

Step 8: Task 4 — Timer

Add:

PYTHON

30 second timer

Display:

TEXT

Time: 29

Time: 28

Stop game when:

TEXT

Time = 0

Add restart:

TEXT

Press R to restart

Commit 4

BASH

```
git add .  
git commit -m "Task 4: Added timer restart and final score"  
git push
```

Step 9: Record AFTER Video (10 sec)

Show:

- ' Click bug fixed
- ' Bonus balloon
- ' Penalty balloon
- ' Lives decreasing
- ' Timer
- ' Game over
- ' Restart

Save:

TEXT

after.mp4

Step 10: Push Everything

Final push:

BASH

```
git push origin main
```

Your Git History Should Look Like This

TEXT

Task 1: Fixed balloon click detection bug

Task 2: Added normal bonus and penalty balloons

Task 3: Added lives system and game over

Task 4: Added timer restart and final score

Prompt You Can Give ChatGPT/Claude For The Lab

Copy this exactly:

TEXT

I have a Pygame Balloon Pop game.

Task 1:

Fix the click detection bug where squared distance is compared to radius.

Task 2:

Add three balloon types:

- Normal (+10 points)
- Bonus (+25 points)
- Penalty (-10 points)

Give each type a different color.

Task 3:

Add 3 lives.

Lose one life when a balloon reaches the bottom.

Game ends at 0 lives.

Task 4:

Add a 30-second timer.

Display time remaining.

Stop game at 0 seconds.

Show final score.

Allow pressing R to restart.

Give complete updated code for each file and explain where to replace code.

This follows exactly what the handout expects: use an LLM, make separate commits, record before/after videos, and upload chat history.

ASSISTANT

Yep— this error is **very simple**. Nothing is wrong with Git yet. PowerShell is just telling you that your Desktop folder isn't at:

TEXT

```
C:\Users\madhu\Desktop
```

Your current location is:

TEXT

```
C:\Users\madhu
```

and `cd Desktop` failed because Windows couldn't find a folder with that name there.

Since you have previously used OneDrive for your Desktop, your Desktop is probably here:

TEXT

```
C:\Users\madhu\OneDrive\Desktop
```

Do this now

Copy-paste this:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop"
```

Then press Enter.

You should see something like:

TEXT

```
(base) PS C:\Users\madhu\OneDrive\Desktop>
```

Then clone your fork

Important: use the URL of **your fork**, not the SETAPESU26 original repository.

For example:

POWERSHELL

```
git clone https://github.com/YOUR-USERNAME/05_balloon_pop.git
```

Replace `YOUR-USERNAME` with your GitHub username.

Then:

POWERSHELL

```
cd 05_balloon_pop
```

Then check:

POWERSHELL

```
git status
```

You should see something like:

TEXT

```
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Don't do anything else yet

Once you run:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop"
```

send me the output/screenshot. I'll take you through the cloning !' running the original game !' recording the **10-second BEFORE video** !' Task 1 !' commit 1, one step at a time.

ASSISTANT

Good — you successfully cloned the repository and entered it. '

The new error is:

TEXT

```
ERROR: Could not open requirements file:
[Errno 2] No such file or directory: 'requirements.txt'
```

That means **the cloned repository does not have `requirements.txt` in its current root folder**. Don't create one yet—we should first see what was actually cloned.

Step 1 — See the files in the repository

Run this **exactly**:

POWERSHELL

```
dir
```

You should get a list containing things like:

TEXT

```
game
main.py
README.md
...
```

Step 2 — Also check the Git status

Run:

POWERSHELL

```
git status
```

Step 3 — If you want to check whether requirements exists anywhere

Run:

POWERSHELL

```
Get-ChildItem -Recurse -Filter requirements.txt
```

Don't install anything or modify any files yet.

Send me the output of those **three commands** (a screenshot is fine). Then I'll tell you exactly what to do next.

Your goal right now is simply:

TEXT

```
C l o n e '
!"
E n t e r   r e p o s i t o r y   '
!"
F i n d   a c t u a l   p r o j e c t   f i l e s   !•   W E   A R E   H E R E
!"
Install dependencies
!"
Run ORIGINAL game
!"
Record 10-second BEFORE video
!"
Task 1 + Commit 1
!"
Task 2 + Commit 2
!"
Task 3 + Commit 3
!"
Task 4 + Commit 4
!"
Record AFTER video
!"
Push final repo
```

Also, your lab specifically says to run the original Python game and record the 10-second video **before making changes**, and to have separate commits for each task.

ASSISTANT

Perfect. Now we know exactly what happened. Ø=ÜM

You cloned the repository, but the actual game is **one folder deeper**.

Right now you're here:

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop
```

Inside it, PowerShell shows:

TEXT

```
balloon-pop
README.md
.gitignore
```

So the actual project is inside:

TEXT

```
05_balloon_pop\balloon-pop
```

That's why `requirements.txt` wasn't found.

Step 1 — Enter the actual game folder

Copy-paste:

POWERSHELL

```
cd balloon-pop
```

Your prompt should become:

TEXT

```
(base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>
```

Step 2 — Check the files

Now run:

POWERSHELL

```
dir
```

You should see the actual project files, probably something like:

TEXT

```
game
main.py
requirements.txt
README.md
```

Step 3 — Install requirements

Only after you confirm `requirements.txt` is there, run:

POWERSHELL

```
pip install -r requirements.txt
```

Step 4 — Run the ORIGINAL game

Then:

POWERSHELL

```
python main.py
```

& pVery important

Do NOT modify any code yet.

The lab specifically requires you to run the original code and record a **10-second BEFORE video before making**

changes.

So our order is:

```
TEXT
cd balloon-pop
! "
dir
! "
pip install -r requirements.txt
! "
python main.py
! "
PLAY ORIGINAL GAME
! "
RECORD 10-SECOND BEFORE VIDEO
! "
ONLY THEN start Task 1
```

Run these **two commands now**:

```
POWERSHELL
cd balloon-pop
dir
```

Send me the screenshot/output. Then I'll tell you the next exact command.

```
USER
how to play the game
```

ASSISTANT

It's very simple — this is a **clicking game**. Ø<ß^

How to play the ORIGINAL game

1. Run:

```
POWERSHELL
python main.py
```

2. A Pygame window called **Balloon Pop** will open.
3. You'll see **pink balloons** appearing at the top of the screen and falling downward. The starter code creates balloons with random sizes and speeds.
4. **Use your mouse and left-click directly on a balloon** before it reaches the bottom.
5. If you successfully pop it:
 - The balloon disappears.
 - Your score increases by **10 points** in the original version.
6. Keep clicking balloons as they fall.

For your BEFORE video

You actually **want to demonstrate the bug**.

Because the original click detection is broken, try this:

First: click exactly near the **center** of a balloon !' it should usually pop.

Then: click closer to the **edge** of a balloon !' it may look like you're clicking inside the balloon, but it won't pop.

That's the bug your Task 1 is asking you to fix. The code currently calculates squared distance but compares it directly with the normal radius.

Also notice that the original game has **no lives and no timer yet**.

What to record for the 10-second BEFORE video

You can literally record something like:

0–3 sec: Show balloons falling.

3–5 sec: Click the center of a balloon !' it pops.

5–8 sec: Click near the visible edge of another balloon !' it doesn't pop.

8–10 sec: Show that there is only a score and no lives/timer.

That's enough to demonstrate the **before state**.

Don't make any code changes yet. Record this video first, because the lab specifically requires the before video before you modify the game.

USER

OKAY I RECORDED THIS NOW WHAT

ASSISTANT

YES. ~~0<10~~ **The BEFORE video is done. Do not change it or delete it.**

Now we start the actual lab work **one task at a time**, and importantly, we need **4 separate commits**, one for each task. The lab explicitly requires separate commits for each task.

We are now doing:

TASK 1 — FIX THE CLICK BUG

Your current code has this bug:

PYTHON

```
distance_squared = dx * dx + dy * dy
if distance_squared <= balloon.radius:
```

The problem is that `distance_squared` is being compared with the normal radius.

STEP 1 — Use this as your LLM prompt

Since the assignment specifically asks you to use an LLM as your pair-programming/debugging partner, **send this prompt in this chat**:

Prompt:

I am working on Task 1 of my Pygame Balloon Pop lab. The current click detection is buggy. In `game/click_detection.py`, the code calculates the squared distance between the mouse click and the balloon center:

```
distance_squared = dx * dx + dy * dy
```

but compares it directly with `balloon.radius`.

Fix this bug so that a click anywhere inside the visible circular balloon correctly registers as a hit. Keep the change minimal and do not implement any of the other tasks yet. Explain what you changed and why.

That's your **LLM interaction for Task 1**.

STEP 2 — Make the change

For Task 1, your `game/click_detection.py` should become:

PYTHON

```
def check_pop(balloons, click_pos):
    """
    Returns the balloon that was clicked, or None if the click missed
    every balloon.
    """
    for balloon in balloons:
        dx = click_pos[0] - balloon.x
        dy = click_pos[1] - balloon.y
        distance_squared = dx * dx + dy * dy

        if distance_squared <= balloon.radius * balloon.radius:
            return balloon

    return None
```

What changed?

Only this:

Before:

PYTHON

```
if distance_squared <= balloon.radius:
```

After:

PYTHON

```
if distance_squared <= balloon.radius * balloon.radius:
```

Because we're comparing:

TEXT

```
distance2
```

with:

TEXT

```
radius2
```

Now the comparison is mathematically correct.

STEP 3 — Test Task 1

Save the file.

Then run:

POWERSHELL

```
python main.py
```

Try clicking balloons:

Test 1

Click near the **center**.

'i|p Balloon should pop.

Test 2

Click near the **edge but still inside the balloon**.

'i|p Balloon should also pop.

Test 3

Click clearly **outside** a balloon.

'i|p Balloon should NOT pop.

If all three work, **Task 1 is complete**.

STEP 4 — Check Git

Close the game and go back to PowerShell.

Run:

POWERSHELL

```
git status
```

You should see something similar to:

TEXT

```
modified: game/click_detection.py
```

STEP 5 — See exactly what changed

Run:

POWERSHELL

```
git diff
```

You should see the change from:

PYTHON

```
if distance_squared <= balloon.radius:
```

to:

PYTHON

```
if distance_squared <= balloon.radius * balloon.radius:
```

This is useful because you're supposed to critically review the LLM's changes rather than blindly accept them.

STEP 6 — COMMIT TASK 1

Once you've tested it:

POWERSHELL

```
git add game/click_detection.py
```

Then:

POWERSHELL

```
git commit -m "Task 1: Fix balloon click detection"
```

Then:

POWERSHELL

```
git push
```

STEP 7 — Verify the commit

Run:

POWERSHELL

```
git log --oneline -5
```

You should see something like:

TEXT

```
abc1234 Task 1: Fix balloon click detection  
xxxxxxx Initial commit
```

🎉 TASK 1 = DONE

& IMPORTANT — DON'T DO TASK 2 YET

We are deliberately doing:

TEXT

```
B E F O R E   V I D E O   '

TASK 1
! "
L L M   p r o m p t   '
! "
Change code
! "
Test
! "
git add
! "
git commit
! "
git push
! "
TASK 1 COMPLETE

T A S K   2   ! •   n e x t
```

The starter project currently has only one balloon type and no lives system/timer, so we don't want to accidentally implement multiple tasks in the same commit.

Do this now:

1. Send the Task 1 prompt above in this chat.

Then I'll act as your LLM pair-programming partner and we'll review the change before you commit it.

USER

I am working on Task 1 of my Pygame Balloon Pop lab. The current click detection is buggy. In `game/click_detection.py`, the code calculates the squared distance between the mouse click and the balloon center:

```
distance_squared = dx * dx + dy * dy
```

but compares it directly with `balloon.radius`.

Fix this bug so that a click anywhere inside the visible circular balloon correctly registers as a hit. Keep the change minimal and do not implement any of the other tasks yet. Explain what you changed and why.

ASSISTANT

Exactly. For **Task 1 only**, make this one minimal change.

Your current code in `game/click_detection.py` is:

PYTHON

```
distance_squared = dx * dx + dy * dy
if distance_squared <= balloon.radius:
    return balloon
```

The problem is that `distance_squared` represents **distance²**, while `balloon.radius` represents just **distance**. The

starter code indeed makes this incorrect comparison.

Corrected code

Replace the contents of `game/click_detection.py` with:

PYTHON

```
def check_pop(balloons, click_pos):  
    """  
    Returns the balloon that was clicked, or None if the click missed  
    every balloon.  
    """  
    for balloon in balloons:  
        dx = click_pos[0] - balloon.x  
        dy = click_pos[1] - balloon.y  
        distance_squared = dx * dx + dy * dy  
  
        if distance_squared <= balloon.radius * balloon.radius:  
            return balloon  
  
    return None
```

Why?

Think of a balloon with:

TEXT

radius = 20

The maximum allowed distance from its center is 20 pixels.

So mathematically:

TEXT

d i s t a n c e ² "d r a d i u s

But we're calculating:

TEXT

distance²

Therefore we need:

TEXT

d i s t a n c e ² "d r a d i u s ²

which in code is:

PYTHON

```
distance_squared <= balloon.radius * balloon.radius
```

This means **any click inside the visible circular balloon** can now register, including clicks near its edge.

Important

We have **NOT** changed:

- balloon types 'L

- score system 'L
- lives 'L
- timer 'L
- restart 'L

So this is cleanly **Task 1 only**.

Now do this on your computer

Open:

TEXT

```
05_balloon_pop
% % %   b a l l o o n - p o p
        % % %   g a m e
                % % %   c l i c k _ d e t e c t i o n . p y
```

Replace the code, **save with Ctrl + S**, then run:

POWERSHELL

```
python main.py
```

Test the balloon near its edge.

If it works, close the game and run:

POWERSHELL

```
git status
```

Then:

POWERSHELL

```
git diff
```

Don't commit yet. Send me the output of `git diff`, and I'll check that you only changed Task 1 before we make **Commit 1**.

USER

give me what to do exactly step by step. i finished recprding now what to do

ASSISTANT

Absolutely. We'll do **one step at a time** so you don't accidentally mix the 4 tasks or commits.

You have already completed:

' Forked the repository ' Cloned it ' Entered `balloon-pop` ' Installed/verified the project ' Ran the original game ' Recorded the **10-second BEFORE video**

The lab says the remaining work is to fix the broken code, add the features one by one, make separate commits, then record the AFTER video and submit the materials.

NOW: TASK 1 ONLY

Step 1 — Open the project folder

You should currently be at:

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

If your PowerShell is somewhere else, run:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop"
```

Step 2 — Open the project in VS Code

Run:

POWERSHELL

```
code .
```

If VS Code opens, great.

If `code .` doesn't work, manually open VS Code and choose:

File !' Open Folder !'

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

Step 3 — Open `click_detection.py`

In VS Code, look at the left **Explorer** panel.

You should have:

TEXT

```
balloon-pop
%
% % %   g a m e
%       % % %   b a l l o o n . p y
%       % % %   c l i c k _ d e t e c t i o n . p y      !•   O P E N   T H I S
%       % % %   g a m e _ e n g i n e . p y
%       % % %   r e n d e r e r . p y
%
% % %   m a i n . p y
% % %   R E A D M E . m d
% % %   r e q u i r e m e n t s . t x t
```

Click:

TEXT

```
game
    !"
click_detection.py
```

The original code contains:

PYTHON

```
distance_squared = dx * dx + dy * dy
if distance_squared <= balloon.radius:
    return balloon
```

That is the bug.

Step 4 — Replace the code

Delete everything inside `click_detection.py`.

Paste this:

PYTHON

```
def check_pop(balloons, click_pos):
    """
    Returns the balloon that was clicked, or None if the click missed
    every balloon.
    """
    for balloon in balloons:
        dx = click_pos[0] - balloon.x
        dy = click_pos[1] - balloon.y
        distance_squared = dx * dx + dy * dy

        if distance_squared <= balloon.radius * balloon.radius:
            return balloon

    return None
```

The **ONLY** important change is:

PYTHON

```
if distance_squared <= balloon.radius * balloon.radius:
```

Instead of:

PYTHON

```
if distance_squared <= balloon.radius:
```

We are **not touching anything else** for Task 1.

Step 5 — Save

Press:

TEXT

Ctrl + S

Step 6 — Test Task 1

Go back to PowerShell.

Make sure you're in:

TEXT

C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop

Run:

POWERSHELL

python main.py

The game opens.

Test it like this:

Test A: Click the middle of a balloon.

'ip It should pop.

Test B: Click close to the edge of a balloon, but still inside it.

'ip It should pop.

Test C: Click outside a balloon.

'ip It should NOT pop.

If all three work:

' TASK 1 IS WORKING.

Close the game.

Step 7 — Check what Git sees

In PowerShell:

POWERSHELL

git status

You should see something like:

TEXT

modified: game/click_detection.py

That's good.

Step 8 — Check the exact change

Now run:

```
POWERSHELL
git diff
```

You should see that the only meaningful change is:

```
DIFF
- if distance_squared <= balloon.radius:
+ if distance_squared <= balloon.radius * balloon.radius:
```

& STOP HERE

Do NOT do `git add`, `git commit`, or Task 2 yet.

Send me a screenshot of your:

```
POWERSHELL
git diff
```

I'll check that Task 1 is correct and then I'll give you the **exact commands for Commit 1 and Push 1**.

Then we'll move to **Task 2**, and repeat the same process. This keeps your required **4 separate commits** clean.

ASSISTANT

Perfect. ' **Task 1 is correct.**

Your `git diff` shows exactly the change we wanted:

```
DIFF
- if distance_squared <= balloon.radius:
+ if distance_squared <= balloon.radius * balloon.radius:
```

Nothing else from the game logic was changed. The `No newline at end of file` message is not a problem, but we can leave it as-is for this lab.

Now let's **commit Task 1 and push it to your GitHub fork**.

STEP 1 — Stage Task 1

In the same PowerShell window, run:

```
POWERSHELL
git add game/click_detection.py
```

There should be no error.

STEP 2 — Check what's staged

Run:

POWERSHELL

```
git status
```

You should now see:

TEXT

```
Changes to be committed:
  modified:   game/click_detection.py
```

If you see that, continue.

STEP 3 — Make Commit 1

Run exactly:

POWERSHELL

```
git commit -m "Task 1: Fix balloon click detection"
```

You should get something similar to:

TEXT

```
[main abc1234] Task 1: Fix balloon click detection
1 file changed, 1 insertion(+), 1 deletion(-)
```

<This is Commit 1 of your required 4 commits.

STEP 4 — Push Commit 1 to GitHub

Run:

POWERSHELL

```
git push origin main
```

You should see something similar to:

TEXT

```
To https://github.com/yourusername/05_balloon_pop.git
xxxxxxx..abc1234  main -> main
```

That means **Task 1 is now on your GitHub fork.**

STEP 5 — Verify

Run:

POWERSHELL

```
git status
```

You want:

TEXT

```
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Then run:

POWERSHELL

```
git log --oneline -3
```

You should see your Task1 commit at the top, something like:

TEXT

```
abc1234 Task 1: Fix balloon click detection
xxxxxxx Initial commit
```

STOP AFTER THIS

Don't start Task2 yet.

Send me a screenshot of the output from:

POWERSHELL

```
git push origin main
```

and:

POWERSHELL

```
git status
```

I'll confirm **Commit 1 is safely pushed**, and then we'll do **Task 2 (Normal + Bonus + Penalty balloons)** with its own LLM prompt and **Commit 2**.

ASSISTANT

Perfect. ' **Commit 1 is successfully completed and pushed.**

Your screenshot confirms everything:

TEXT

```
481d52d Task 1: Fix balloon click detection
```

And:

TEXT

```
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

So we have:

Task	Status	Commit
Task 1 – Fix click detection	' D O N E	481d52d
Task 2 – Balloon types	#ó N e x t	—
Task 3 – Lives	#ó	—
Task 4 – Timer	#ó	—

The push also went to your personal fork:

TEXT

https://github.com/MadhumithaDhondi/05_balloon_pop.git

Now TASK 2 — Balloon Types Ø<ß^

Do not change anything yet. First we need to interact with the LLM, because your assignment specifically wants the tasks developed iteratively with an LLM.

Step 1 — Send this prompt

Copy-paste this **exact prompt** into this chat:

TEXT

I have completed Task 1 of my Pygame Balloon Pop lab and committed it separately.

Now I need to implement Task 2 only.

Task 2:

Introduce at least three balloon types:

1. Normal balloon - regular points (+10)
2. Bonus balloon - more points (+25)
3. Penalty balloon - reduces the score (-10)

Each balloon type must have its own clearly distinguishable color.

When a balloon is popped, the correct score effect must happen based on its type.

Please inspect the existing project structure and code I provided. Make the minimum necessary changes to implement ONLY Task 2.

Do not implement the lives system or timer yet.

Explain:

1. Which files need to change
2. What you changed
3. How the balloon type is represented
4. How the score changes for each type
5. Why your implementation works

Why we're doing it this way

Your starter `GameEngine` currently creates a `Balloon` with just position, radius and speed, and gives every popped balloon the same `POINTS_PER_POP = 10`.

The `Balloon` class already has a `color` property, so we'll likely be able to build the visual distinction around that existing structure.

But don't start editing files from my explanation yet.

Send the prompt above, and I'll give you the exact Task 2 code changes **file by file**, then we'll test them, check `git diff`, and make **Commit 2** separately.

USER

I have completed Task 1 of my Pygame Balloon Pop lab and committed it separately.

Now I need to implement Task 2 only.

Task 2: Introduce at least three balloon types:

1. Normal balloon - regular points (+10)
2. Bonus balloon - more points (+25)
3. Penalty balloon - reduces the score (-10)

Each balloon type must have its own clearly distinguishable color.

When a balloon is popped, the correct score effect must happen based on its type.

Please inspect the existing project structure and code I provided. Make the minimum necessary changes to implement **ONLY** Task 2.

Do not implement the lives system or timer yet.

Explain:

1. Which files need to change
2. What you changed
3. How the balloon type is represented
4. How the score changes for each type
5. Why your implementation works

ASSISTANT

Perfect. **Task 2 only**. We will NOT touch lives or timer.

Based on your starter code, the `GameEngine` currently creates every balloon the same way and awards every popped balloon `10` points. Your `Balloon` class already has a `color` attribute, which makes it easy to distinguish the types visually.

What we're going to change

Only **2 files**:

TEXT

```
game/
% % %   b a l l o o n . p y           !•   a d d   b a l l o o n   t y p e   +   p o i n t s
% % %   g a m e _ e n g i n e . p y   !•   r a n d o m l y   c r e a t e   t y p e s   +   u s e   t
```

We will **not** change:

TEXT

```
c l i c k _ d e t e c t i o n . p y 'L
m a i n . p y 'L
r e n d e r e r . p y 'L
l i v e s 'L
```

```
t i m e r 'L
```

STEP 1 — Open `balloon.py`

Open:

TEXT

```
g a m e !'   b a l l o o n . p y
```

You currently have a constructor like:

PYTHON

```
def __init__(self, x, y, radius, speed, color=(220, 90, 120)):
```

We are going to give each balloon a **type** and automatically assign its color and points.

Replace the contents of `balloon.py` with:

PYTHON

```
import pygame

class Balloon:
    def __init__(self, x, y, radius, speed, balloon_type="normal"):
        self.x = x
        self.y = y
        self.radius = radius
        self.speed = speed
        self.balloon_type = balloon_type

        if balloon_type == "normal":
            self.color = (220, 90, 120)
            self.points = 10
        elif balloon_type == "bonus":
            self.color = (80, 200, 100)
            self.points = 25
        elif balloon_type == "penalty":
            self.color = (220, 60, 60)
            self.points = -10

    def update(self):
        self.y += self.speed

    def is_past_bottom(self, height):
        return self.y - self.radius > height

    def get_rect(self):
        return pygame.Rect(
            int(self.x - self.radius),
            int(self.y - self.radius),
            self.radius * 2,
            self.radius * 2,
        )
```

What did we add?

This:

PYTHON

```
self.balloon_type = balloon_type
```

stores whether the balloon is:

TEXT

```
normal  
bonus  
penalty
```

Then:

PYTHON

```
self.color
```

controls what it looks like.

And:

PYTHON

```
self.points
```

controls what happens to the score.

So:

Type	Color	Points
Normal	Pink	+10
Bonus	Green	+25
Penalty	Red	-10

STEP 2 — Open `game_engine.py`

Open:

TEXT

```
game !' game_engine.py
```

Your current code has:

PYTHON

```
POINTS_PER_POP = 10
```

and when you pop a balloon:

PYTHON

```
self.score += POINTS_PER_POP
```

That's why every balloon currently gives the same score.

We need to change that.

Replace your `game_engine.py` with:

PYTHON

```
import random

from game.balloon import Balloon
from game.click_detection import check_pop
from game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES = 45

class GameEngine:
    def __init__(self):
        self.balloons = []
        self.frames_until_spawn = 0
        self.score = 0

    def _spawn_balloon(self):
        radius = random.randint(16, 44)
        x = random.randint(radius + 10, WIDTH - radius - 10)
        speed = random.uniform(1.5, 3.0)

        balloon_type = random.choice(["normal", "bonus", "penalty"])

        self.balloons.append(
            Balloon(
                x=x,
                y=-radius,
                radius=radius,
                speed=speed,
                balloon_type=balloon_type,
            )
        )

    def handle_click(self, pos):
        popped = check_pop(self.balloons, pos)

        if popped is not None:
            self.balloons.remove(popped)
            self.score += popped.points

    def update(self):
        self.frames_until_spawn -= 1

        if self.frames_until_spawn <= 0:
            self._spawn_balloon()
            self.frames_until_spawn = SPAWN_INTERVAL_FRAMES

        for b in self.balloons:
            b.update()

        self.balloons = [
            b for b in self.balloons
            if not b.is_past_bottom(HEIGHT)
        ]

    def draw(self, surface, font):
        from game import renderer

        renderer.draw_scene(surface, self.balloons)
        renderer.draw_text(
            surface,
```

```
        font,
        f"Score: {self.score}",
        (10, 10),
    )
```

STEP 3 — Understand the important changes

1. Random balloon type

We added:

PYTHON

```
balloon_type = random.choice(["normal", "bonus", "penalty"])
```

Every time a balloon spawns, the game randomly chooses one of the three types.

2. We pass the type into the Balloon

PYTHON

```
Balloon(
    x=x,
    y=-radius,
    radius=radius,
    speed=speed,
    balloon_type=balloon_type,
)
```

So the Balloon knows whether it's normal, bonus or penalty.

3. Each balloon stores its own points

For example:

PYTHON

```
normal !' self.points = 10
bonus !' self.points = 25
penalty !' self.points = -10
```

4. The score now uses the balloon's points

Instead of:

PYTHON

```
self.score += POINTS_PER_POP
```

we now have:

PYTHON

```
self.score += popped.points
```

This is the important part.

If you pop:

TEXT

```
N o r m a l ! '   s c o r e   + 1 0  
B o n u s ! '   s c o r e   + 2 5  
P e n a l t y ! '   s c o r e   - 1 0
```

STEP 4 — Save both files

In VS Code:

TEXT

Ctrl + S

Make sure both are saved:

TEXT

```
game/balloon.py  
game/game_engine.py
```

STEP 5 — Run the game

Go to PowerShell.

You should still be here:

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

Run:

POWERSHELL

```
python main.py
```

STEP 6 — Test Task 2

You need to verify **all three types**.

Play for a little while until you see different colored balloons.

Pink balloon

Click it.

Expected:

TEXT

```
Score +10
```

Green balloon

Click it.

Expected:

TEXT

Score +25

Red balloon

Click it.

Expected:

TEXT

Score -10

For example, if your score is:

TEXT

Score: 35

and you pop a red penalty balloon:

TEXT

Score: 25

That's correct.

STEP 7 — Important: Don't worry about missed balloons yet

You might notice balloons reaching the bottom without anything happening.

That's okay.

Task 3 will add the lives/miss system.

We are intentionally doing **Task 2 only** right now.

STEP 8 — Check Git BEFORE committing

Close the game.

Run:

POWERSHELL

git status

You should see:

TEXT

modified: game/balloon.py
modified: game/game_engine.py

Then:

```
POWERSHELL
git diff
```

Check that the changes are only related to:

- balloon types
- colors
- points
- random type selection
- score based on balloon type

& STOP HERE

Do not commit yet.

Send me a screenshot of:

```
POWERSHELL
git status
```

and:

```
POWERSHELL
git diff
```

I'll check that you haven't accidentally changed Task3 or Task4.

Once it's correct, we'll do:

```
TEXT
Task 2
!"
git add
!"
Commit 2
!"
git push
!"
verify GitHub
```

Then we'll move on to **Task 3 — 3 lives**.

ASSISTANT

Yes — I can see your `git diff`. ~~One~~ **The Task 2 changes in `balloon.py` look correct.**

However, **don't commit yet**. Your screenshot is only showing part of the diff, and you're currently inside Git's diff viewer.

1. Exit the `git diff` screen

At the bottom you can see:

TEXT

:

That means Git has opened the diff in a pager.

Press:

TEXT

q

Just the letter **q**.

You should return to:

TEXT

```
(base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>
```

2. Check the Task 2 files

Run:

POWERSHELL

```
git status
```

You should see:

TEXT

```
modified: game/balloon.py
modified: game/game_engine.py
```

3. Important: Test the game first

Before committing, run:

POWERSHELL

```
python main.py
```

You need to verify Task 2 actually works.

You should now see **three different balloon colors**:

- ~~Ø~~ Pink = Normal = **+10**
- ~~Ø~~ Green = Bonus = **+25**
- ~~Ø~~ Red = Penalty = **-10**

Click them and watch the score.

For example:

TEXT

Score: 0

Click pink:

TEXT

Score: 10

Click green:

TEXT

Score: 35

Click red:

TEXT

Score: 25

You don't need to get these exact numbers in this order; that's just an example.

Also check:

If you miss a balloon and it reaches the bottom, **nothing should happen to lives**, because we haven't implemented Task 3 yet.

That's correct.

4. Close the game

Once you've confirmed the three types work, close the Pygame window.

Then run:

POWERSHELL

```
git status
```

5. Check the complete diff

Now run:

POWERSHELL

```
git diff
```

If it opens the pager again, you can scroll through it and then press:

TEXT

```
q
```

What we want to see

Changes related to:

TEXT

```
balloon.py
    balloon_type
    colors
    points

game_engine.py
    random balloon type
    popped.points
```

We DON'T want:

TEXT

```
lives
timer
restart
game over
```

because those belong to Tasks 3 and 4.

One small thing I noticed

Your screenshot shows the `get_rect()` formatting changed:

DIFF

```
- return pygame.Rect(
-     int(self.x - self.radius), int(self.y - self.radius),
-     self.radius * 2, self.radius * 2,
- )
+ return pygame.Rect(
+     int(self.x - self.radius),
+     int(self.y - self.radius),
+     self.radius * 2,
+     self.radius * 2,
+ )
```

That's **not functionally different** — it is just formatting. It's fine, although ideally Task 2 should keep unrelated changes minimal.

So your immediate next action is:

Press ``q` !` run:

POWERSHELL

```
git status
```

Then run:

POWERSHELL

```
python main.py
```

Test the **pink, green and red balloons**.

Tell me whether you see all 3 colors and whether the scores change correctly. **Don't commit yet.**

USER

im seeing 3 colours

ASSISTANT

Perfect! ~~That means the visual part of Task 2 is working.~~ '

Now we need to check the **score effects** before we commit.

Step 1 — Test the pink balloon

Click a **pink** balloon.

It should give:

TEXT

+10

For example:

TEXT

Score: 0

!' p i n k

!' S c o r e : 1 0

Step 2 — Test the green balloon

Click a **green** balloon.

It should give:

TEXT

+25

For example:

TEXT

Score: 10

!' g r e e n

!' S c o r e : 3 5

Step 3 — Test the red balloon

Click a **red** balloon.

It should give:

TEXT

-10

For example:

TEXT

```
Score: 35
!'  r e d
!'  S c o r e :    2 5
```

Important

You **do not need to worry about lives yet**. If a balloon reaches the bottom, that's expected because **Task 3 hasn't been implemented**.

If all three scores work

Close the game and run:

POWERSHELL

```
git status
```

Then:

POWERSHELL

```
git diff
```

If everything looks good, **don't commit yet**—send me the result of `git status` (and the diff if it shows anything unexpected).

Then I'll give you the exact commands for **Commit 2 + Push 2**.

USER

```
yes all good
```

ASSISTANT

Great. ' **Task 2 is fully working**.

Now we need to create **Commit 2**. This is important because your lab requires a separate commit for each task.

STEP 1 — Check status

In PowerShell, run:

POWERSHELL

```
git status
```

You should see something like:

TEXT

```
Changes not staged for commit:
  modified: game/balloon.py
  modified: game/game_engine.py
```

STEP 2 — Stage the Task 2 files

Run:

```
POWERSHELL
git add game/balloon.py game/game_engine.py
```

STEP 3 — Check what you're about to commit

Run:

```
POWERSHELL
git status
```

You should now see:

```
TEXT
Changes to be committed:
  modified: game/balloon.py
  modified: game/game_engine.py
```

If that's what you see, continue.

STEP 4 — Commit Task 2

Run:

```
POWERSHELL
git commit -m "Task 2: Add balloon types and scoring"
```

You should get something like:

```
TEXT
[main XXXXXXX] Task 2: Add balloon types and scoring
2 files changed...
```

STEP 5 — Push to GitHub

Run:

```
POWERSHELL
git push origin main
```

You should see:

TEXT

```
main -> main
```

STEP 6 — Verify

Run:

POWERSHELL

```
git status
```

You want:

TEXT

```
Your branch is up to date with 'origin/main'.
```

```
nothing to commit, working tree clean
```

Then:

POWERSHELL

```
git log --oneline -4
```

You should now have something similar to:

TEXT

```
XXXXXXX Task 2: Add balloon types and scoring
481d52d Task 1: Fix balloon click detection
c9a5e46 Initial Balloon Pop starter
```

Commit 2 is done.

Your progress

Task	Status
Before video	'
Task 1 – Click bug	' Commit 1
Task 2 – 3 balloon types	' Commit 2
Task 3 – 3 lives	#0 Next
Task 4 – 30-sec timer	#0
After video	#0

Run the commands above. Once `git status` says `nothing to commit, working tree clean`, send me the screenshot/output. Then we'll start Task 3 — the 3-life system.

USER

done all good proceed with task 3

ASSISTANT

Absolutely. **Task 3 = Lives/Miss System.** 'dp Ø=Ý%

We're going to keep this completely separate from Task 4. **Do not add the timer yet.**

Your lab requires:

- Start with **3 lives**
- Balloon reaching the bottom = **lose 1 life**
- Popping a balloon = **never loses a life**
- Display remaining lives
- Game ends when lives reach 0

Your current starter engine removes balloons that reach the bottom but doesn't do anything when that happens.

TASK 3 — EXACT STEPS

Step 1 — First, use the LLM prompt

Since your assignment wants you to use an LLM for each feature, send this:

TEXT

I have completed Task 1 and Task 2 of my Pygame Balloon Pop lab, and each task has its own Git commit.

Now I need to implement Task 3 only.

Task 3 requirements:

- Add 3 lives at the start of the game.
- Whenever a balloon reaches the bottom of the screen without being popped, the player loses exactly 1 life.
- Popping a balloon must NEVER cost a life, regardless of whether it is a normal, bonus, or penalty balloon.
- Display the remaining lives on screen.
- End the game when lives reach 0.
- Balloons should no longer be actively playable once the game has ended.

Please inspect the existing project structure and the current Task 1 and Task 2 implementation.

Make the minimum necessary changes for ONLY Task 3.

Do NOT implement the 30-second timer, restart system, or any Task 4 functionality yet.

Explain:

1. Which files need to change.
2. How the lives are stored.
3. How a missed balloon is detected.
4. How the game knows when lives reach zero.
5. How you ensure popping a balloon never removes a life.

Step 2 — What we're expecting

The important logic should basically be:

```
TEXT
Start
    !"
Lives = 3
    !"
Balloon falls
    !"
Did player pop it?
    % % % % % % % % % % % % % % %
YES          NO
    !"          !"
Remove balloon  Reaches bottom
                !"
                Lives - 1
                !"
                Lives = 0?
                /      \
            NO      YES
                !"          !"
            Continue  Game Over
```

Step 3 — Files that will probably change

Based on your current project, the main file that needs modification is:

```
TEXT
game/game_engine.py
```

Your renderer may also need a small change if the existing text-drawing setup needs it.

We should not need to touch:

```
TEXT
game/click_detection.py
game/balloon.py
```

because those already handle Tasks 1 and 2.

Your `main.py` currently sends mouse clicks to `engine.handle_click()` and continuously calls `engine.update()` and `engine.draw()`.

Step 4 — Don't edit yet

Send the **Task 3 prompt above** to me first.

I'll give you the exact code to put in your files, just like we did for Task 2.

Then we'll do:

```
TEXT
Task 3 code
    !"
Run game
    !"
```

```
Test 3 lives
! "
Test missed balloon
! "
Test popping doesn't lose life
! "
Test game over at 0
! "
git diff
! "
Commit 3
! "
git push
```

Do not add the timer yet. That will be **Task 4** after Commit 3.

USER

I have completed Task 1 and Task 2 of my Pygame Balloon Pop lab, and each task has its own Git commit.

Now I need to implement Task 3 only.

Task 3 requirements:

- Add 3 lives at the start of the game.
- Whenever a balloon reaches the bottom of the screen without being popped, the player loses exactly 1 life.
- Popping a balloon must NEVER cost a life, regardless of whether it is a normal, bonus, or penalty balloon.
- Display the remaining lives on screen.
- End the game when lives reach 0.
- Balloons should no longer be actively playable once the game has ended.

Please inspect the existing project structure and the current Task 1 and Task 2 implementation.

Make the minimum necessary changes for ONLY Task 3. Do NOT implement the 30-second timer, restart system, or any Task 4 functionality yet.

Explain:

1. Which files need to change.
2. How the lives are stored.
3. How a missed balloon is detected.
4. How the game knows when lives reach zero.
5. How you ensure popping a balloon never removes a life.

ASSISTANT

Perfect. **Task 3 only.** We will leave the timer and restart completely untouched.

Based on the project structure and the Task 2 version we've just completed, the cleanest implementation is to change `game/game_engine.py` only.

Your current engine already detects balloons that have gone past the bottom using `is_past_bottom(HEIGHT)` and then removes them. We'll change that removal so that a missed balloon also costs one life.

Step 1 — Open `game/game_engine.py`

In VS Code:

TEXT

```
game
%% %      g a m e _ e n g i n e . p y
```

Step 2 — Replace the entire file

Since you want the exact step-by-step approach, replace the contents of `game_engine.py` with this:

PYTHON

```
import random

from game.balloon import Balloon
from game.click_detection import check_pop
from game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES = 45

class GameEngine:
    def __init__(self):
        self.balloons = []
        self.frames_until_spawn = 0
        self.score = 0
        self.lives = 3
        self.game_over = False

    def _spawn_balloon(self):
        if self.game_over:
            return

        radius = random.randint(16, 44)
        x = random.randint(radius + 10, WIDTH - radius - 10)
        speed = random.uniform(1.5, 3.0)

        balloon_type = random.choice(["normal", "bonus", "penalty"])

        self.balloons.append(
            Balloon(
                x=x,
                y=-radius,
                radius=radius,
                speed=speed,
                balloon_type=balloon_type,
            )
        )

    def handle_click(self, pos):
        if self.game_over:
            return

        popped = check_pop(self.balloons, pos)

        if popped is not None:
            self.balloons.remove(popped)
            self.score += popped.points

    def update(self):
        if self.game_over:
            return
```

```

self.frames_until_spawn -= 1

if self.frames_until_spawn <= 0:
    self._spawn_balloon()
    self.frames_until_spawn = SPAWN_INTERVAL_FRAMES

for b in self.balloons:
    b.update()

remaining_balloons = []

for b in self.balloons:
    if b.is_past_bottom(HEIGHT):
        self.lives -= 1
    else:
        remaining_balloons.append(b)

self.balloons = remaining_balloons

if self.lives <= 0:
    self.lives = 0
    self.game_over = True

def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.balloons)

    renderer.draw_text(
        surface,
        font,
        f"Score: {self.score}",
        (10, 10),
    )

    renderer.draw_text(
        surface,
        font,
        f"Lives: {self.lives}",
        (10, 40),
    )

    if self.game_over:
        renderer.draw_text(
            surface,
            font,
            "GAME OVER",
            (WIDTH // 2 - 70, HEIGHT // 2),
        )

```

Step 3 — Understand what changed

There are **4 important changes**.

\$`We start with 3 lives

Inside `__init__`:

```

PYTHON
self.lives = 3

```

So when the game starts:

TEXT

Lives: 3

\$We added a game-over state

PYTHON

```
self.game_over = False
```

This is simply a yes/no variable:

TEXT

```
False = game is running  
True  = game is over
```

\$Missing a balloon costs one life

Previously, the game essentially did:

TEXT

```
Balloon reaches bottom  
! "  
Remove balloon
```

Now it does:

TEXT

```
Balloon reaches bottom  
! "  
self.lives -= 1  
! "  
Remove balloon
```

This happens here:

PYTHON

```
for b in self.balloons:  
    if b.is_past_bottom(HEIGHT):  
        self.lives -= 1  
    else:  
        remaining_balloons.append(b)
```

So every balloon that reaches the bottom costs exactly one life.

\$Popping does NOT cost a life

This is important.

When you click:

PYTHON

```
if popped is not None:
    self.balloons.remove(popped)
    self.score += popped.points
```

There is **no**:

PYTHON

```
self.lives -= 1
```

Therefore:

TEXT

```
Pop normal      !' score +10 !' lives unchanged
Pop bonus       !' score +25 !' lives unchanged
Pop penalty     !' score -10 !' lives unchanged
```

Exactly what Task 3 requires.

Step 4 — Game ends at 0 lives

This code checks:

PYTHON

```
if self.lives <= 0:
    self.lives = 0
    self.game_over = True
```

So:

TEXT

```
Lives: 3
      !"
miss
      !"
Lives: 2
      !"
miss
      !"
Lives: 1
      !"
miss
      !"
Lives: 0
      !"
GAME OVER
```

We explicitly set:

PYTHON

```
self.lives = 0
```

so you won't accidentally display:

TEXT

Lives: -1

Step 5 — Game stops after Game Over

We added this to `update()`:

PYTHON

```
if self.game_over:
    return
```

That means the game stops updating once you reach zero lives.

We also added this to `_spawn_balloon()`:

PYTHON

```
if self.game_over:
    return
```

So no new balloons are spawned after game over.

And this to `handle_click()`:

PYTHON

```
if self.game_over:
    return
```

So you can't keep popping balloons after the game has ended.

Step 6 — Save

Press:

TEXT

Ctrl + S

Step 7 — Run the game

Go to PowerShell:

TEXT

C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop

Run:

POWERSHELL

python main.py

Step 8 — Test Task 3 properly

We need to test **three things**.

TEST A — Lives are displayed

Immediately when the game starts, you should see something like:

TEXT

Score: 0

Lives: 3

' Good.

TEST B — Missing a balloon costs a life

Don't click one of the balloons.

Let it fall all the way down.

You should see:

TEXT

Lives: 3

then after it reaches the bottom:

TEXT

Lives: 2

Let another one fall:

TEXT

Lives: 1

Let another one fall:

TEXT

Lives: 0

GAME OVER

' Good.

TEST C — Popping does NOT cost a life

This one is important.

Start again by closing and running the game again:

POWERSHELL

python main.py

You should have:

TEXT

Lives: 3

Pop a balloon.

The score should change, but:

TEXT

Lives: 3

should remain.

Try different types too:

TEXT

```
P i n k      !'   s c o r e   + 1 0 !'   l i v e s   u n c h a n g e d
G r e e n    !'   s c o r e   + 2 5 !'   l i v e s   u n c h a n g e d
R e d        !'   s c o r e   - 1 0 !'   l i v e s   u n c h a n g e d
```

' If that happens, Task 3 is working correctly.

Step 9 — Test Game Over

Let three balloons reach the bottom without clicking them.

At zero:

TEXT

Lives: 0
GAME OVER

Then try clicking a balloon if one somehow remains.

Nothing should happen.

Also verify that **new balloons aren't spawning**.

Step 10 — STOP HERE

Once you've tested everything, **do NOT commit yet**.

Close the game and run:

POWERSHELL

git status

Then:

POWERSHELL

git diff

We want Task 3 changes only.

You should see changes primarily in:

TEXT

```
game/game_engine.py
```

And we **do not want** anything involving:

TEXT

```
30-second timer  
restart  
R key  
countdown
```

because those are Task 4.

Your workflow right now

TEXT

```
Task 3 code  
! "  
Save  
! "  
python main.py  
! "  
Check Lives: 3  
! "  
M i s s ! ' L i v e s d e c r e a s e s  
! "  
P o p ! ' L i v e s s t a y s s a m e  
! "  
3 m i s s e s ! ' G A M E O V E R  
! "  
Close game  
! "  
git status  
! "  
git diff
```

Do these tests now. Once they all work, send me a screenshot of `git status`/`git diff`. I'll check it before you make **Commit 3**.

USER

```
make the commit
```

ASSISTANT

Yes. If you already tested Task 3 and everything works, let's make **Commit 3** now.

You should be in:

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

1. Check status

```
POWERSHELL
git status
```

You should see `game/game_engine.py` modified.

2. Stage it

```
POWERSHELL
git add game/game_engine.py
```

3. Commit Task 3

```
POWERSHELL
git commit -m "Task 3: Add lives and game over"
```

You should see something like:

```
TEXT
[main XXXXXXX] Task 3: Add lives and game over
```

4. Push it to your fork

```
POWERSHELL
git push origin main
```

5. Verify

```
POWERSHELL
git status
```

You want:

```
TEXT
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Then:

```
POWERSHELL
git log --oneline -5
```

Your history should now look roughly like:

```
TEXT
XXXXXXX Task 3: Add lives and game over
XXXXXXX Task 2: Add balloon types and scoring
481d52d Task 1: Fix balloon click detection
XXXXXXX Initial commit
```

Commit 3 is now done.

Do not start Task 4 until you confirm the push succeeded. After that, we'll implement the **30-second timer + restart**, make **Commit 4**, then record your final 10-second video.

USER

done proceed

ASSISTANT

Perfect. **Task 3 is done. Now we proceed to Task 4 — the final feature.**

We will do it carefully and **Task 4 gets its own separate commit.**

TASK 4 — 30-SECOND TIMER + RESTART

What Task 4 must add

Your game will now have:

- **30-second countdown**
- **Existing 3 lives**
- **Existing 3 balloon types**
- **Existing scoring**
- **Game ends when:**
 - timer reaches **0**, **OR**
 - lives reach **0**
- **After game ends:**
 - no new balloons spawn
 - clicks don't pop balloons
- **Final score remains visible**
- **Press R to start a completely new round**
 - Score !' **0**
 - Lives !' **3**
 - Timer !' **30**
 - Game starts again

STEP 1 — Open your project

You should already be in:

POWERSHELL

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

If you're not sure, run:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop"
```

Then:

POWERSHELL

```
code .
```

STEP 2 — Give your coding AI this prompt

Use the same AI/Vibe Coding tool you have been using.

Copy-paste this entire prompt:

TEXT

Task 4: Add a 30-second countdown timer and restart functionality.

Build on the current code exactly as it exists after Task 3.

Requirements:

1. Add a 30-second countdown timer.
2. Display the remaining time clearly on the game screen.
3. The timer must be based on real elapsed time, not simply assuming every frame takes exactly the same amount of time.
4. The game should end when either:
 - the timer reaches 0, OR
 - lives reach 0.
5. When the game ends:
 - stop spawning new balloons
 - stop accepting balloon clicks
 - stop active gameplay
 - keep the final score visible
 - clearly display GAME OVER
6. Add a way to start a new round by pressing the R key.
7. When R is pressed after the game ends, reset:
 - score to 0
 - lives to 3
 - timer to 30 seconds
 - balloons to an empty list
 - game_over state
 - spawning state
8. The new round should start normally after pressing R.
9. Do not remove or break any existing functionality:
 - Task 1 click detection fix
 - Task 2 normal/bonus/penalty balloons
 - Task 2 scoring
 - Task 3 lives
 - Task 3 game over
10. Do not add unrelated features.
11. Make the minimum necessary changes.
12. Keep the code beginner-friendly and readable.

Because restarting requires keyboard input, modify main.py if necessary to handle the R key and call an appropriate reset method in GameEngine.

After making the changes, explain exactly which files were changed and provide the complete final code for each changed file.

STEP 3 — VERY IMPORTANT

The AI will probably modify:

TEXT

```
game/game_engine.py  
main.py
```

It should not need to modify:

TEXT

```
game/balloon.py  
game/click_detection.py
```

unless there is a genuine reason.

After it makes the changes, **do not immediately commit**.

First run the game.

STEP 4 — Run the game

In PowerShell:

POWERSHELL

```
python main.py
```

You should see the game.

Test 1 — Timer

Immediately look at the screen.

You should see something like:

TEXT

```
Score: 0  
Lives: 3  
Time: 30
```

The timer should decrease:

TEXT

```
30  
29  
28  
27  
...
```

It should **not** stay at 30.

STEP 5 — Test scoring still works

Pop a normal balloon.

You should get:

TEXT
+10

Pop a bonus balloon:

TEXT
+25

Pop a penalty balloon:

TEXT
-10

Make sure the score still changes correctly.

STEP 6 — Test lives

Let balloons fall without clicking them.

You should see:

TEXT
Lives: 3

then:

TEXT
Lives: 2

then:

TEXT
Lives: 1

then:

TEXT
Lives: 0

At 0:

TEXT
GAME OVER

should appear.

STEP 7 — Test timer game over

This one is important.

Start another round if necessary, then **don't intentionally lose all the lives**.

Wait until the timer reaches:

TEXT

0

The game should end.

You should see something like:

TEXT

Score: 35

Lives: 2

Time: 0

GAME OVER

Press R to restart

The **exact wording may differ**, but the behavior must be correct.

After timer reaches 0:

Try clicking a balloon.

It should **not be popped**.

Also, no new balloons should spawn.

STEP 8 — Test RESTART

Press:

TEXT

R

You should get a fresh game:

TEXT

Score: 0

Lives: 3

Time: 30

and balloons should start spawning again.

This is extremely important because restart is one of the Task 4 requirements.

STEP 9 — Check that Task 3 wasn't broken

After restarting, deliberately let balloons escape again.

Lives should decrease normally:

TEXT

```
3 !' 2 !' 1 !' 0
```

Then game over.

Press **R**.

It should return to:

TEXT

```
Score: 0  
Lives: 3  
Time: 30
```

STEP 10 — Stop the game

Once you've tested everything, close the Pygame window.

Do NOT commit yet if anything is wrong.

STEP 11 — Check Git

In PowerShell:

POWERSHELL

```
git status
```

You should see changes in the files modified for Task 4.

Probably:

TEXT

```
modified:  game/game_engine.py  
modified:  main.py
```

Then run:

POWERSHELL

```
git diff
```

Look through it.

You want to make sure the changes are **only Task 4 changes**.

& STOP HERE

Do these things first:

1. Give the AI the Task 4 prompt.
2. Let it modify the code.
3. Run:

POWERSHELL

```
python main.py
```

4. Test:

- timer
- scoring
- lives
- timer reaching 0
- game over
- clicking after game over
- **R** restart
- reset to 0 score / 3 lives / 30 seconds

5. Then run:

POWERSHELL

```
git status  
git diff
```

Send me the output/screenshot of `git diff` (or the AI's Task 4 code) before committing.

I'll check that Task 4 is correct, then I'll give you the **exact Commit 4 commands**, and after that we'll do the **AFTER video + final GitHub verification + chat-history deliverable**.

USER

Task 4: Add a 30-second countdown timer and restart functionality.

Build on the current code exactly as it exists after Task 3.

Requirements:

1. Add a 30-second countdown timer.
2. Display the remaining time clearly on the game screen.
3. The timer must be based on real elapsed time, not simply assuming every frame takes exactly the same amount of time.
4. The game should end when either:
 - the timer reaches 0, OR
 - lives reach 0.
5. When the game ends:
 - stop spawning new balloons
 - stop accepting balloon clicks
 - stop active gameplay
 - keep the final score visible
 - clearly display GAME OVER
6. Add a way to start a new round by pressing the R key.
7. When R is pressed after the game ends, reset:
 - score to 0
 - lives to 3

- timer to 30 seconds
- balloons to an empty list
- game_over state
- spawning state

8. The new round should start normally after pressing R.

9. Do not remove or break any existing functionality:

- Task 1 click detection fix
- Task 2 normal/bonus/penalty balloons
- Task 2 scoring
- Task 3 lives
- Task 3 game over

10. Do not add unrelated features.

11. Make the minimum necessary changes.

12. Keep the code beginner-friendly and readable.

Because restarting requires keyboard input, modify main.py if necessary to handle the R key and call an appropriate reset method in GameEngine.

After making the changes, explain exactly which files were changed and provide the complete final code for each changed file.

ASSISTANT

Yes— **that is the correct Task 4 prompt.** Ø=ÜM

Now **paste that prompt into your coding/Vibe Coding AI** (the same one you used for Tasks 1–3).

After it modifies the files

Do **not** commit yet.

First run:

POWERSHELL

```
python main.py
```

Then test these **in this exact order**:

1. Timer

- Starts at about 30
- Counts down using real time.

2. Scoring

- Normal = +10
- Bonus = +25
- Penalty = -10

3. Lives

- Starts at 3
- Missed balloon !' 2 !' 1 !' 0

4. Game over from lives

- At 0 lives, GAME OVER appears.

- Balloons cannot be popped anymore.
- New balloons stop spawning.

5. Restart

- Press **R**.
- Confirm:

TEXT

```
Score: 0
Lives: 3
Time: 30
```

- New balloons start spawning.

6. Game over from timer

- Start another round.
- Keep enough lives.
- Wait until timer reaches 0.
- Confirm game ends and final score remains visible.

7. Restart again

- Press **R**.
- Confirm everything resets.

Then close the game and run:

POWERSHELL

```
git status
```

and:

POWERSHELL

```
git diff
```

Send me the `git diff` output (or screenshots of it) here.

I'll check the Task 4 implementation before we make **Commit 4**. After that we'll do the **AFTER 10-second video and final submission checks**.

USER

Task 4: Add a 30-second countdown timer and restart functionality.

Build on the current code exactly as it exists after Task 3.

Requirements:

1. Add a 30-second countdown timer.
2. Display the remaining time clearly on the game screen.
3. The timer must be based on real elapsed time, not simply assuming every frame takes exactly the same amount of time.
4. The game should end when either:
 - the timer reaches 0, OR

- lives reach 0.
- 5. When the game ends:
 - stop spawning new balloons
 - stop accepting balloon clicks
 - stop active gameplay
 - keep the final score visible
 - clearly display GAME OVER
- 6. Add a way to start a new round by pressing the R key.
- 7. When R is pressed after the game ends, reset:
 - score to 0
 - lives to 3
 - timer to 30 seconds
 - balloons to an empty list
 - game_over state
 - spawning state
- 8. The new round should start normally after pressing R.
- 9. Do not remove or break any existing functionality:
 - Task 1 click detection fix
 - Task 2 normal/bonus/penalty balloons
 - Task 2 scoring
 - Task 3 lives
 - Task 3 game over
- 10. Do not add unrelated features.
- 11. Make the minimum necessary changes.
- 12. Keep the code beginner-friendly and readable.

Because restarting requires keyboard input, modify main.py if necessary to handle the R key and call an appropriate reset method in GameEngine.

After making the changes, explain exactly which files were changed and provide the complete final code for each changed file.

ASSISTANT

Yes— **use that exact prompt** in your coding/Vibe Coding AI. It is the Task 4 prompt we need.

Once the AI finishes modifying the code, **don't commit yet**.

Run:

```
POWERSHELL
python main.py
```

Test:

- Timer starts at **30** and counts down.
- Normal balloon = **+10**
- Bonus balloon = **+25**
- Penalty balloon = " **1 0**

- Lives start at **3** and decrease when balloons are missed.
- Game ends at **0 lives**.
- Game also ends when timer reaches **0**.
- After game over, balloons cannot be popped and new balloons stop spawning.
- Press **R** !' everything resets:
 - Score = 0
 - Lives = 3
 - Timer = 30
 - New balloons spawn again.

Then close the game and run:

POWERSHELL

```
git status
```

POWERSHELL

```
git diff
```

Send me the output of `git diff` (or screenshots of the changed code).

I'll check it before we make **Commit 4**.

USER

Task 4: Add a 30-second countdown timer and restart functionality.

Build on the current code exactly as it exists after Task 3.

Requirements:

1. Add a 30-second countdown timer.
2. Display the remaining time clearly on the game screen.
3. The timer must be based on real elapsed time, not simply assuming every frame takes exactly the same amount of time.
4. The game should end when either:
 - the timer reaches 0, OR
 - lives reach 0.
5. When the game ends:
 - stop spawning new balloons
 - stop accepting balloon clicks
 - stop active gameplay
 - keep the final score visible
 - clearly display GAME OVER
6. Add a way to start a new round by pressing the R key.
7. When R is pressed after the game ends, reset:
 - score to 0
 - lives to 3
 - timer to 30 seconds

- balloons to an empty list
- game_over state
- spawning state

8. The new round should start normally after pressing R.

9. Do not remove or break any existing functionality:

- Task 1 click detection fix
- Task 2 normal/bonus/penalty balloons
- Task 2 scoring
- Task 3 lives
- Task 3 game over

10. Do not add unrelated features.

11. Make the minimum necessary changes.

12. Keep the code beginner-friendly and readable.

Because restarting requires keyboard input, modify main.py if necessary to handle the R key and call an appropriate reset method in GameEngine.

After making the changes, explain exactly which files were changed and provide the complete final code for each changed file.

ASSISTANT

Yes— **that is the Task 4 prompt**. Paste it into your Vibe Coding AI now.

Once it finishes changing the code, **don't commit yet**.

Run:

POWERSHELL

```
python main.py
```

Test these:

1. Timer starts at **30** and counts down.
2. Normal = **+10**, bonus = **+25**, penalty = " **1.0**
3. Lives start at **3** and decrease when balloons are missed.
4. Game ends when lives reach **0**.
5. Game ends when timer reaches **0**.
6. After game over, clicking does nothing and no new balloons spawn.
7. Press **R** and verify:

TEXT

```
Score: 0
Lives: 3
Time: 30
```

8. Verify balloons start spawning again.

Then close the game and run:

POWERSHELL

```
git status
```

then:

POWERSHELL

```
git diff
```

Send me the `git diff` output or screenshots of the changed code.

I'll check Task 4 before you make the **fourth and final commit**.

USER

can you give me the changed codes

ASSISTANT

Yes. Since you want the **actual changed code**, use these two files for Task 4.

1. game/game_engine.py

Replace the **entire contents** of `game/game_engine.py` with:

PYTHON

```
import random
import pygame

from game.balloon import Balloon
from game.click_detection import check_pop
from game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES = 45
GAME_DURATION_SECONDS = 30

class GameEngine:
    def __init__(self):
        self.balloons = []
        self.frames_until_spawn = 0
        self.score = 0
        self.lives = 3
        self.game_over = False

        self.start_time = pygame.time.get_ticks()
        self.time_remaining = GAME_DURATION_SECONDS

    def _spawn_balloon(self):
        if self.game_over:
            return

        radius = random.randint(16, 44)
        x = random.randint(radius + 10, WIDTH - radius - 10)
        speed = random.uniform(1.5, 3.0)

        balloon_type = random.choice(["normal", "bonus", "penalty"])
```

```

        self.balloons.append(
            Balloon(
                x=x,
                y=-radius,
                radius=radius,
                speed=speed,
                balloon_type=balloon_type,
            )
        )

    def handle_click(self, pos):
        if self.game_over:
            return

        popped = check_pop(self.balloons, pos)

        if popped is not None:
            self.balloons.remove(popped)
            self.score += popped.points

    def update(self):
        if self.game_over:
            return

        elapsed_seconds = (pygame.time.get_ticks() - self.start_time) / 1000
        self.time_remaining = max(
            0,
            GAME_DURATION_SECONDS - int(elapsed_seconds)
        )

        if self.time_remaining <= 0:
            self.time_remaining = 0
            self.game_over = True
            self.balloons = []
            return

        self.frames_until_spawn -= 1

        if self.frames_until_spawn <= 0:
            self._spawn_balloon()
            self.frames_until_spawn = SPAWN_INTERVAL_FRAMES

        for b in self.balloons:
            b.update()

        remaining_balloons = []

        for b in self.balloons:
            if b.is_past_bottom(HEIGHT):
                self.lives -= 1
            else:
                remaining_balloons.append(b)

        self.balloons = remaining_balloons

        if self.lives <= 0:
            self.lives = 0
            self.game_over = True
            self.balloons = []

    def reset(self):
        self.balloons = []
        self.frames_until_spawn = 0
        self.score = 0
        self.lives = 3

```

```

self.game_over = False

self.start_time = pygame.time.get_ticks()
self.time_remaining = GAME_DURATION_SECONDS

def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.balloons)

    renderer.draw_text(
        surface,
        font,
        f"Score: {self.score}",
        (10, 10),
    )

    renderer.draw_text(
        surface,
        font,
        f"Lives: {self.lives}",
        (10, 40),
    )

    renderer.draw_text(
        surface,
        font,
        f"Time: {self.time_remaining}",
        (10, 70),
    )

    if self.game_over:
        renderer.draw_text(
            surface,
            font,
            "GAME OVER",
            (WIDTH // 2 - 70, HEIGHT // 2),
        )

        renderer.draw_text(
            surface,
            font,
            f"Final Score: {self.score}",
            (WIDTH // 2 - 90, HEIGHT // 2 + 35),
        )

        renderer.draw_text(
            surface,
            font,
            "Press R to restart",
            (WIDTH // 2 - 100, HEIGHT // 2 + 70),
        )

```

2. main.py

Replace the **entire contents** of `main.py` with:

```

PYTHON

import pygame

from game.game_engine import GameEngine
from game.renderer import WIDTH, HEIGHT

```

```

def main():
    pygame.init()

    screen = pygame.display.set_mode((WIDTH, HEIGHT))
    pygame.display.set_caption("Balloon Pop")

    font = pygame.font.Font(None, 36)
    clock = pygame.time.Clock()

    engine = GameEngine()

    running = True

    while running:
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                running = False

            elif event.type == pygame.MOUSEBUTTONDOWN:
                engine.handle_click(event.pos)

            elif event.type == pygame.KEYDOWN:
                if event.key == pygame.K_r and engine.game_over:
                    engine.reset()

        engine.update()
        engine.draw(screen, font)

        pygame.display.flip()
        clock.tick(60)

    pygame.quit()

if __name__ == "__main__":
    main()

```

Why these two files changed

game/game_engine.py

Task 4 adds:

- 30 second duration
- Real elapsed-time calculation using `pygame.time.get_ticks()`
- `time_remaining`
- Timer display
- Timer-based game over
- `reset()` function
- Final score display
- Restart instructions
- Clearing balloons when game ends

main.py

Task 4 adds keyboard handling:

PYTHON

```
elif event.type == pygame.KEYDOWN:  
    if event.key == pygame.K_r and engine.game_over:  
        engine.reset()
```

So **R** only restarts after the game has ended.

Now do this

Save both files.

Then run:

POWERSHELL

```
python main.py
```

First check

You should see:

TEXT

```
Score: 0  
Lives: 3  
Time: 30
```

Wait a few seconds and confirm it becomes:

TEXT

```
Time: 29
```

then:

TEXT

```
Time: 28
```

etc.

Then test R

Let the game end, then press:

TEXT

```
R
```

It should reset to:

TEXT

```
Score: 0  
Lives: 3  
Time: 30
```

Don't commit yet. After testing, run:

POWERSHELL

```
git diff
```

and send me the output. I'll verify it before we make **Task 4's final commit**.

USER

why is it stopping at 18 seconds

ASSISTANT

Ah — **this is actually working correctly**. Your screenshot tells us exactly what happened.

Look at your game:

TEXT

```
Score: 150
Lives: 0
Time: 18
GAME OVER
```

Why did it stop at 18?

Because **you lost all 3 lives before the 30 seconds were over**.

Your Task 4 requirement says the game ends when **either**:

Timer reaches 0 OR lives reach 0.

So in your case:

TEXT

```
30 seconds
! "
you play
! "
lives: 3 !' 2 !' 1 !' 0
! "
GAME OVER
! "
timer was still 18 seconds
```

Once `Lives = 0`, the game stops, so the timer stays at **18**.

This is actually a good test '

You have demonstrated:

- ' Score works: 150
- ' Lives work: 3 !' 0
- ' Game Over works
- ' Final Score is displayed
- ' Timer was running
- ' Game stopped when lives reached 0

Now we need to test the OTHER game-over condition

We need to make sure the game can also end because of the **timer**.

Step 1

Press:

TEXT

R

It should reset to:

TEXT

Score: 0

Lives: 3

Time: 30

Step 2

Now **pop balloons continuously** so you don't lose all your lives.

Basically, keep clicking balloons.

Wait until:

TEXT

Time: 0

You should then see:

TEXT

Score: [whatever you achieved]

Lives: [probably 1-3]

Time: 0

GAME OVER

Final Score: [same score]

Press R to restart

Step 3

Try clicking a balloon after **GAME OVER**.

It should **not pop**.

Step 4

Press **R**.

It should reset to:

TEXT

Score: 0

Lives: 3

Time: 30

One important thing

Your screenshot also shows:

TEXT

```
git status
modified: game/game_engine.py
```

That's **correct** right now. We haven't committed Task 4 yet.

Don't commit yet.

First test the **timer !' 0 !' GAME OVER !' R restart** sequence.

If that works, tell me **"timer works"**, and I'll give you the exact commands for **Task 4 Commit 4 + push**.

USER

pygame 2.6.1 (SDL 2.28.4, Python 3.13.5) Hello from the pygame community.

<https://www.pygame.org/contribute.html> (base) PS

C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff diff --git

a/balloon-pop/game/game_engine.py b/balloon-pop/game/game_engine.py index e181a81..cd66b99 100644 ---

a/balloon-pop/game/game_engine.py +++ b/balloon-pop/game/game_engine.py @@ -1,4 +1,5 @@ import random
+import pygame

from game.balloon import Balloon from game.click_detection import check_pop @@ -6,6 +7,7 @@ from
game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES= 45 +GAME_DURATION_SECONDS= 30

class GameEngine: @@ -16,6 +18,9 @@ class GameEngine: self.lives = 3 self.game_over = False

- self.start_time = pygame.time.get_ticks()
- self.time_remaining = GAME_DURATION_SECONDS

def _spawn_balloon(self): if self.game_over: return @@ -50,6 +55,18 @@ class GameEngine: if self.game_over:
return

- elapsed_seconds = (pygame.time.get_ticks() - self.start_time) / 1000
- self.time_remaining = max(
 - 0,
 - GAME_DURATION_SECONDS- int(elapsed_seconds)
 -)
- if self.time_remaining <= 0:
- self.time_remaining = 0
- self.game_over = True
- self.balloons = []
- return

self.frames_until_spawn -= 1

: diff --git a/balloon-pop/game/game_engine.py b/balloon-pop/game/game_engine.py index e181a81..cd66b99
100644 --- a/balloon-pop/game/game_engine.py +++ b/balloon-pop/game/game_engine.py @@ -1,4 +1,5 @@
import random +import pygame

```

from game.balloon import Balloon from game.click_detection import check_pop @@ -6,6 +7,7 @@ from
game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES= 45 +GAME_DURATION_SECONDS= 30

class GameEngine: @@ -16,6 +18,9 @@ class GameEngine: self.lives = 3 self.game_over = False
    • self.start_time = pygame.time.get_ticks()
    • self.time_remaining = GAME_DURATION_SECONDS
def _spawn_balloon(self): if self.game_over: return @@ -50,6 +55,18 @@ class GameEngine: if self.game_over:
return

    • elapsed_seconds = (pygame.time.get_ticks() - self.start_time) / 1000
    • self.time_remaining = max(
    • 0,
    • GAME_DURATION_SECONDS- int(elapsed_seconds)
    • )
    • if self.time_remaining <= 0:
    • self.time_remaining = 0
    • self.game_over = True
    • self.balloons = []
: diff --git a/balloon-pop/game/game_engine.py b/balloon-pop/game/game_engine.py index e181a81..cd66b99
100644 --- a/balloon-pop/game/game_engine.py +++ b/balloon-pop/game/game_engine.py @@ -1,4 +1,5 @@
import random +import pygame

from game.balloon import Balloon from game.click_detection import check_pop @@ -6,6 +7,7 @@ from
game.renderer import WIDTH, HEIGHT

SPAWN_INTERVAL_FRAMES= 45 +GAME_DURATION_SECONDS= 30 this is the output next what to do

```

ASSISTANT

Yes. You're at the **Git diff check**, and I can already see one important thing.

Your `git diff` is showing the Task 4 changes, but it is currently inside Git's **pager** — that's why you see `:` at the bottom and the diff appears repeated/truncated.

Step 1 — Exit the diff screen

Press:

```

TEXT
q

```

on your keyboard.

You should return to:

```

TEXT
PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>

```

Step 2 — Check exactly which files changed

Run:

POWERSHELL

```
git status
```

Then run:

POWERSHELL

```
git diff --stat
```

We **expect** Task 4 to have changed:

TEXT

```
game/game_engine.py  
main.py
```

Your output currently only shows:

TEXT

```
game/game_engine.py
```

So I want to verify whether `main.py` was actually changed.

Step 3 — Check `main.py`

Run:

POWERSHELL

```
git diff -- main.py
```

If it shows nothing, that's a problem because we need the `R` key restart functionality.

You should see something similar to:

PYTHON

```
elif event.type == pygame.KEYDOWN:  
    if event.key == pygame.K_r and engine.game_over:  
        engine.reset()
```

Step 4 — Check the complete game-engine diff

Run:

POWERSHELL

```
git diff -- game/game_engine.py
```

If it opens the pager again, just press:

TEXT

```
q
```

after looking at it.

But DON'T commit yet

Before committing, we need to make sure `main.py` contains the R-key code.

So do these **three commands** now:

POWERSHELL

```
git status
```

POWERSHELL

```
git diff --stat
```

POWERSHELL

```
git diff -- main.py
```

Send me the output of those.

Then I'll tell you exactly what to do next. **Don't run `git add` or `git commit` yet.**

USER

```
self.lives = 0 (base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git status
On branch main
Your branch is up to date with 'origin/main'.
```

```
Changes not staged for commit: (use "git add <file>..." to update what will be committed) (use "git restore <file>..." to
discard changes in working directory) modified:   game/game_engine.py modified:   main.py
```

```
no changes added to commit (use "git add" and/or "git commit -a") (base) PS
```

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff --stat balloon-pop/game/game_engine.py |
49 ++++++ balloon-pop/main.py | 25 ++++++----- 2
```

```
files changed, 62 insertions(+), 12 deletions(-) (base) PS
```

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>
```

ASSISTANT

Perfect. ' This is exactly what we expected.

Your `git status` confirms **only the two Task 4 files changed**:

TEXT

```
game/game_engine.py
main.py
```

And your `git diff --stat` confirms:

TEXT

```
2 files changed
```

So now we just need to make sure the restart code is actually present in `main.py`.

Step 1 — Run this:

POWERSHELL

```
git diff -- main.py
```

Look for this part:

PYTHON

```
elif event.type == pygame.KEYDOWN:
    if event.key == pygame.K_r and engine.game_over:
        engine.reset()
```

If you see it, **tell me "yes"** and we'll move directly to the final commit.

If you **don't** see it, send me what `git diff -- main.py` shows and I'll fix it before committing.

Don't commit yet.

USER

```
(base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff -- main.py diff --git
a/balloon-pop/main.py b/balloon-pop/main.py index 2a0d793..e207bb0 100644 --- a/balloon-pop/main.py +++
b/balloon-pop/main.py @@ -1,33 +1,34 @@ -""" -Balloon Pop (Lab Starter) - -Run with: python3 main.py - -Click
balloons to pop them before they reach the bottom. -""" - import pygame

from game.game_engine import GameEngine -from game.renderer import WINDOW_SIZE +from game.renderer
import WIDTH, HEIGHT

def main(): pygame.init()

    • screen = pygame.display.set_mode(WINDOW_SIZE)
+
    • screen = pygame.display.set_mode((WIDTH,HEIGHT))
pygame.display.set_caption("BalloonPop") +
    • font = pygame.font.Font(None, 36)
clock = pygame.time.Clock()

    • font = pygame.font.SysFont("consolas", 22)
engine = GameEngine() + running = True + while running: for event in pygame.event.get(): if event.type ==
pygame.QUIT: running = False + elif event.type == pygame.MOUSEBUTTONDOWN:
engine.handle_click(event.pos)

    • elif event.type == pygame.KEYDOWN:
    • if event.key == pygame.K_r and engine.game_over:
    • engine.reset()
+ engine.update() engine.draw(screen, font)
:
```

ASSISTANT

Yes!' The important part is there:

PYTHON

```
elif event.type == pygame.KEYDOWN:
    if event.key == pygame.K_r and engine.game_over:
        engine.reset()
```

So **R restart is implemented correctly**.

However, I noticed something we should fix **before committing**. Your `main.py` changed some things that Task 4 did **not** require:

- Removed the original file documentation
- Changed `WINDOW_SIZE` !' (WIDTH, HEIGHT)
- Changed the font from `SysFont("consolas", 22)` !' `Font(None, 36)`

The assignment asks for minimum necessary changes, so let's clean that up. We want the diff to show **only the R-key functionality** in `main.py`.

Step 1 — Replace `main.py`

Open `main.py` and replace everything with:

PYTHON

```
"""
Balloon Pop (Lab Starter)

Run with: python3 main.py

Click balloons to pop them before they reach the bottom.
"""

import pygame

from game.game_engine import GameEngine
from game.renderer import WINDOW_SIZE

def main():
    pygame.init()
    screen = pygame.display.set_mode(WINDOW_SIZE)
    pygame.display.set_caption("Balloon Pop")
    clock = pygame.time.Clock()
    font = pygame.font.SysFont("consolas", 22)

    engine = GameEngine()
    running = True

    while running:
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                running = False

            elif event.type == pygame.MOUSEBUTTONDOWN:
                engine.handle_click(event.pos)

            elif event.type == pygame.KEYDOWN:
                if event.key == pygame.K_r and engine.game_over:
                    engine.reset()

        engine.update()
        engine.draw(screen, font)

        pygame.display.flip()
```

```
        clock.tick(60)

    pygame.quit()

if __name__ == "__main__":
    main()
```

Why?

Now the only functional addition to the original `main.py` is:

```
PYTHON
elif event.type == pygame.KEYDOWN:
    if event.key == pygame.K_r and engine.game_over:
        engine.reset()
```

That's exactly what we need.

Step 2 — Save it

Press:

```
TEXT
Ctrl + S
```

Step 3 — Run the game again

```
POWERSHELL
python main.py
```

Test:

Timer

You should see approximately:

```
TEXT
Score: 0
Lives: 3
Time: 30
```

Game over

Let the lives reach 0 or wait until timer reaches 0.

Restart

After `GAME OVER`, press:

```
TEXT
R
```

It should reset:

TEXT

Score: 0
Lives: 3
Time: 30

Step 4 — Check the final diff

Close the game and run:

POWERSHELL

```
git diff --stat
```

Then:

POWERSHELL

```
git diff -- main.py
```

Now the `main.py` diff should be very small — essentially just the `KEYDOWN` block.

Then:

POWERSHELL

```
git status
```

STOP HERE

Do not commit yet.

Send me the new `git diff -- main.py` output. Once I confirm it, we'll make **Task 4 Commit 4**.

USER

```
ello from the pygame community. https://www.pygame.org/contribute.html (base) PS
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff --stat balloon-pop/game/game_engine.py |
49 ++++++ balloon-pop/main.py | 8 +++++- 2 files changed,
56 insertions(+), 1 deletion(-) (base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff --
main.py diff --git a/balloon-pop/main.py b/balloon-pop/main.py index 2a0d793..2f2085c 100644 ---
a/balloon-pop/main.py +++ b/balloon-pop/main.py @@ -21,13 +21,19 @@ def main():

engine = GameEngine() running = True + while running: for event in pygame.event.get(): if event.type ==
pygame.QUIT: running = False + elif event.type == pygame.MOUSEBUTTONDOWN:
engine.handle_click(event.pos)

• elif event.type == pygame.KEYDOWN:
• if event.key == pygame.K_r and engine.game_over:
• engine.reset()
+ engine.update() engine.draw(screen, font)

@@ -38,4 +44,4 @@ def main():

if name == "main":
```

- main()
- main()

\ No newline at end of file (base) PS C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop>git diff --main.py

ASSISTANT

Perfect. ' **The Task 4 diff now looks clean.**

Your `main.py` has only the necessary functional addition:

PYTHON

```
elif event.type == pygame.KEYDOWN:
    if event.key == pygame.K_r and engine.game_over:
        engine.reset()
```

The extra changes are just blank-line formatting plus the missing newline at the end of the file. That's harmless.

Your summary:

TEXT

```
2 files changed
56 insertions
1 deletion
```

is reasonable because most of the Task 4 work is in `game/game_engine.py`.

One thing before committing

You already tested that **lives reaching 0** causes game over.

Now make sure you have also tested:

30 !' 29 !' ... !' 0 !' GAME OVER !' press R !' 30, 3 lives, score 0

If that works, we're ready.

TASK 4 — FINAL COMMIT

Run these **one at a time**:

1. Check status

POWERSHELL

```
git status
```

You should see:

TEXT

```
modified: game/game_engine.py
modified: main.py
```

2. Stage both Task 4 files

POWERSHELL

```
git add game/game_engine.py main.py
```

3. Verify staged changes

POWERSHELL

```
git status
```

You should now see:

TEXT

```
Changes to be committed:
  modified: game/game_engine.py
  modified: main.py
```

4. Make the fourth commit

POWERSHELL

```
git commit -m "Task 4: Add timer and restart"
```

5. Push it to your GitHub fork

POWERSHELL

```
git push origin main
```

6. Final status

POWERSHELL

```
git status
```

You want:

TEXT

```
nothing to commit, working tree clean
```

and:

TEXT

```
Your branch is up to date with 'origin/main'
```

7. Verify all four commits

POWERSHELL

```
git log --oneline -5
```

You should see approximately:

TEXT

```
Task 4: Add timer and restart
Task 3: Add lives and game over
```

Task 2: Add balloon types and scoring
Task 1: Fix balloon click detection
Initial Balloon Pop starter

Do these commands now. Once the push succeeds, send me the `git log --oneline -5` output. Then we'll do the **AFTER video**, and finally check the remaining submission deliverables.

USER

all perfect

ASSISTANT

Perfect. ~~0<11~~ The coding portion is **DONE**.

You now have all **4 separate commits** and the final code pushed to your GitHub fork.

What remains for the assignment

You need to finish the **submission/deliverables**:

1. ' Task 1 — click detection fix
2. ' Task 2 — balloon types + scoring
3. ' Task 3 — lives + game over
4. ' Task 4 — 30-second timer + restart
5. ' Separate Git commits
6. ' Code pushed to your personal fork
7. #6 **AFTER video — 10 seconds**
8. #6 **Chat history exported as DOC/PDF**
9. #6 **Final submission/check**

The assignment specifically requires before/after videos, updated code, and exported chat history, with separate commits for each task.

~~0<11~~ NOW RECORD THE AFTER VIDEO

You already have your **BEFORE video**, so don't record that again.

For the **AFTER video**, make a short ~10-second demonstration showing as many features as possible.

Recommended sequence

Start the game and show:

0–2 sec

Show:

TEXT

Score: 0
Lives: 3
Time: 30

and balloons appearing.

2–5 sec

Pop different balloon types.

Show the score changing.

5–7 sec

Let a balloon escape so the lives decrease.

7–10 sec

If possible, show the timer and/or game-over/restart behavior.

You don't need to perfectly demonstrate every feature in 10 seconds—the video is just evidence that the completed game works.

Important

Don't spend forever trying to get the perfect video. ~~0=~~0=

Your code and commits are the main evidence.

0=0=When we need the CHAT HISTORY

The assignment requires:

Chat history exported as doc/pdf

So you need to preserve the conversation showing your Vibe Coding prompts and the changes.

If your Vibe Coding tool has an **Export / Download chat** option, export it as **PDF** if available.

If it only gives you text/Markdown, that's also fine—we can turn it into a proper PDF/DOC.

Your final checklist

Before submitting, you should have:

Requirement	Status
Before video	'
Task 1	'
Task 2	'
Task 3	'
Task 4	'
Separate commits	'
GitHub push	'
After video	#0

Requirement	Status
Chat history PDF/DOC	#6
No PR to original repo	'

The assignment explicitly says **do not raise a PR to the main/original repository**.

So your next action is simply:

Record the 10-second AFTER video.

Then tell me **"after video done"**, and I'll walk you through the **chat-history export + final submission checklist step by step**.

USER

after video done

ASSISTANT

Perfect. **Your actual project work is complete.**

Now let's finish the **submission part** so you don't miss anything.

Final submission checklist

You should have these **4 things**:

1. BEFORE video '

You already recorded the required 10-second video **before making changes**.

2. AFTER video '

Done just now.

3. Updated GitHub repository '

Your fork should contain:

TEXT

Task 1: Fix balloon click detection
Task 2: Add balloon types and scoring
Task 3: Add lives and game over
Task 4: Add timer and restart

And you already pushed the final version.

4. Chat history export #6

This is the **last thing we need to do**.

The assignment asks for the **chat history to be exported as a DOC/PDF** along with the videos and updated code.

Chat History

If your Vibe Coding tool has an option such as:

- **Export**
- **Download conversation**
- **Export chat**
- **Save as PDF**

use that.

Best format

PDF is easiest.

Name it something like:

TEXT

PES1UG24AM923_Lab4_Chat_History.pdf

Keep the chat showing:

- Task 1 prompt
- Task 1 changes
- Task 2 prompt
- Task 2 changes
- Task 3 prompt
- Task 3 changes
- Task 4 prompt
- Task 4 changes
- Testing/fixes

That's useful evidence of the Vibe Coding process.

Your final files

I recommend naming your submission files:

TEXT

PES1UG24AM923_Lab4_Before.mp4

PES1UG24AM923_Lab4_After.mp4

PES1UG24AM923_Lab4_Chat_History.pdf

And your GitHub repository contains the updated project.

One last GitHub check

If you want to be **100% safe**, run:

POWERSHELL

git status

It should say:

TEXT

nothing to commit, working tree clean

Then:

POWERSHELL

```
git log --oneline -5
```

You should have the four Task commits.

🎉 You're essentially finished.

The assignment's required workflow was: clone/use the assigned repo, run the starter, record the before video, make the fixes/features one by one, record the after video, push the updated code, and maintain separate commits for each task.

You have completed that workflow. '

The only remaining action is **exporting/saving the chat history**, then submit the three files + GitHub repository according to whatever submission portal your instructor provided.

USER

so in my github i want to push the videos how to do that

ASSISTANT

Yes! You can push the **BEFORE and AFTER videos directly into your GitHub repository**. Since your repo is already cloned and connected, the easiest way is through PowerShell.

& pOne thing first: GitHub has a **100 MB limit for normal files**. If either video is over 100 MB, don't try to push it normally—we'll use Git LFS or another method.

Step 1 — Find your videos

You need the actual locations of:

- BEFORE video
- AFTER video

For example, they might be in:

TEXT

Downloads

or:

TEXT

Videos

Let's first locate them.

In PowerShell, run:

POWERSHELL

```
Get-ChildItem "$HOME\Downloads" -File | Where-Object {$_.Extension -match '\.(mp4|mov|webm)$'} |  
Select-Object Name, Length, FullName
```

Also check Videos:

POWERSHELL

```
Get-ChildItem "$HOME\Videos" -File | Where-Object {$_.Extension -match '\.(mp4|mov|webm)$'} |  
Select-Object Name, Length, FullName
```

You'll get something like:

TEXT

Name	Length	FullName
before.mp4	8500000	C:\Users\madhu\Downloads\before.mp4
after.mp4	9200000	C:\Users\madhu\Downloads\after.mp4

Step 2 — Create a folder in your project

Go to your project:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop"
```

Create a folder called `videos`:

POWERSHELL

```
mkdir videos
```

Step 3 — Copy the videos into it

For example, if your videos are in **Downloads**, use:

POWERSHELL

```
Copy-Item "$HOME\Downloads\before.mp4" ".\videos\Lab4_Before.mp4"
```

and:

POWERSHELL

```
Copy-Item "$HOME\Downloads\after.mp4" ".\videos\Lab4_After.mp4"
```

& Replace `before.mp4` and `after.mp4` with your **actual filenames**.

Then check:

POWERSHELL

```
Get-ChildItem .\videos
```

You should see:

TEXT

```
Lab4_Before.mp4
Lab4_After.mp4
```

Step 4 — Check video sizes

Run:

POWERSHELL

```
Get-ChildItem .\videos | Select-Object Name,
@{Name="SizeMB";Expression={[math]::Round($_.Length/1MB,2)}}
```

If both are under 100 MB

Great. Continue normally.

If either is over 100 MB

STOP. Don't `git add` yet.

Tell me the sizes and I'll show you the Git LFS method.

Step 5 — Add the videos to Git

If they're both under 100 MB:

POWERSHELL

```
git add videos
```

Then:

POWERSHELL

```
git status
```

You should see:

TEXT

Changes to be committed:

```
new file: videos/Lab4_Before.mp4
new file: videos/Lab4_After.mp4
```

Step 6 — Commit the videos

Since your four Task commits are already complete, **don't mix the videos into Task 4.**

Make a separate commit:

POWERSHELL

```
git commit -m "Add Lab 4 before and after videos"
```

Step 7 — Push

POWERSHELL

```
git push origin main
```

Wait for it to finish.

You should get something like:

TEXT

```
main -> main
```

Step 8 — Check GitHub

Open your repository:

You should now have:

TEXT

```
05_balloon_pop
%
% % %   g a m e /
% % %   v i d e o s /
%       % % %   L a b 4 _ B e f o r e . m p 4
%       % % %   L a b 4 _ A f t e r . m p 4
% % %   m a i n . p y
% % %   r e q u i r e m e n t s . t x t
% % %   R E A D M E . m d
```

& Important

Don't upload the videos through GitHub's normal **Add file !' Upload files** if you're already working from your cloned repository. It's much cleaner to put them in your local `videos` folder, commit them, and push them.

First run the two commands from Step 1 and send me the output. I'll tell you the exact `Copy-Item` commands using your actual video filenames.

USER

theyre not in dowbloads theyre in SE LAB can i just copy path of them or msth

ASSISTANT

Yes! **Copying the file paths is actually the easiest way.** Ø=ÜM

Since your videos are in your **SE LAB** folder, we can copy them into your GitHub project's `videos` folder using their

paths.

Do this:

1. Open the SE LAB folder in File Explorer

Find your **BEFORE** video.

Right-click it !' **Copy as path**.

It will give you something like:

TEXT

```
"C:\Users\madhu\OneDrive\Desktop\SE LAB\Before.mp4"
```

Do the same for the **AFTER** video.

2. Send me the two paths here

Just paste:

TEXT

BEFORE:

```
"C:\Users\madhu\OneDrive\Desktop\SE LAB\...mp4"
```

AFTER:

```
"C:\Users\madhu\OneDrive\Desktop\SE LAB\...mp4"
```

Don't move or rename anything yet.

I'll give you the exact copy-paste commands to put both videos into:

TEXT

```
05_balloon_pop\balloon-pop\videos\
```

and then we'll push them to GitHub.

USER

```
okay here the path for before and after in order "C:\Users\madhu\OneDrive\Desktop\SE LAB\lab4_before.mp4"
```

```
"C:\Users\madhu\OneDrive\Desktop\SE LAB\lab4_after.mp4"
```

ASSISTANT

Perfect. ~~0=0~~ We have both exact paths.

You're currently working from:

TEXT

```
C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop
```

We'll copy both videos into a `videos` folder, then commit and push them.

Step 1 — Go to your GitHub project

Run:

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop"
```

Step 2 — Create the videos folder

POWERSHELL

```
mkdir videos
```

If it says the folder already exists, that's completely fine.

Step 3 — Copy the BEFORE video

Copy-paste:

POWERSHELL

```
Copy-Item "C:\Users\madhu\OneDrive\Desktop\SE LAB\lab4_before.mp4" ".\videos\lab4_before.mp4"
```

Step 4 — Copy the AFTER video

POWERSHELL

```
Copy-Item "C:\Users\madhu\OneDrive\Desktop\SE LAB\lab4_after.mp4" ".\videos\lab4_after.mp4"
```

Step 5 — Make sure they're there

Run:

POWERSHELL

```
Get-ChildItem .\videos
```

You should see:

TEXT

```
lab4_before.mp4  
lab4_after.mp4
```

Step 6 — CHECK FILE SIZES & p

Before adding them to Git, run:

POWERSHELL

```
Get-ChildItem .\videos | Select-Object Name,  
@{Name="SizeMB";Expression={[math]::Round($_.Length/1MB,2)}}
```

You'll get something like:

TEXT

Name	SizeMB
lab4_before.mp4	8.42
lab4_after.mp4	9.17

If BOTH are below 100 MB

You're good. **Don't do anything else yet.**

Send me that output and I'll give you the exact commands to:

TEXT

```
git add
!' c o m m i t
!' p u s h
!' v e r i f y   o n   G i t H u b
```

If either is above 100 MB

STOP. Don't run `git add` yet.

Send me the sizes and I'll give you the Git LFS commands instead.

USER

all good theyre below 10 mb

ASSISTANT

Perfect! ' Since both videos are **under 10 MB**, we can upload them normally—no Git LFS needed.

You're ready to add them to Git.

1. Check that you're in the project folder

POWERSHELL

```
cd "C:\Users\madhu\OneDrive\Desktop\05_balloon_pop\balloon-pop"
```

2. Add the videos

POWERSHELL

```
git add videos
```

3. Check what will be committed

POWERSHELL

```
git status
```

You should see:

TEXT

Changes to be committed:

```
new file: videos/lab4_before.mp4
new file: videos/lab4_after.mp4
```

4. Commit them

POWERSHELL

```
git commit -m "Add Lab 4 before and after videos"
```

5. Push to GitHub

POWERSHELL

```
git push origin main
```

Wait for it to finish successfully.

6. Verify

POWERSHELL

```
git status
```

You want:

TEXT

```
nothing to commit, working tree clean
```

Then open your repository:

You should see:

TEXT

```
videos/
  lab4_before.mp4
  lab4_after.mp4
```

Do these commands in order. If `git status` shows exactly those two video files staged, you're safe to commit and push.

USER

what is this vibe coding thing i used ths chat to vibe code

ASSISTANT

Yes— **you used this ChatGPT conversation itself as your Vibe Coding tool.** You do **not** need some separate “Vibe Coding” website/tool.

In your lab, **Vibe Coding** basically means:

You describe what you want the code to do in natural language !' AI suggests/changes the code !' you run and test it !' you ask AI to fix/improve it !' you commit the changes.

That is exactly what we did for your Balloon Pop project.

What you did

You started with broken/starter code and used me to make changes through prompts:

1. Task 1:

You told me to fix the balloon click detection bug. !' We changed the distance calculation.

2. Task 2:

You prompted me to add:

- Normal balloons !' +10
- Bonus balloons !' +25
- Penalty balloons !' " 1 0

3. Task 3:

You prompted me to add:

- 3 lives
- Losing a life when a balloon reaches the bottom
- Game Over at 0 lives

4. Task 4:

You prompted me to add:

- 30-second timer
- Game Over when timer reaches 0
- Final score
- **R** to restart
- Resetting the game

And you **tested each change yourself** and made **4 separate Git commits**, which is exactly the workflow the lab asks for.

The lab specifically says to use a Vibe Coding tool to fix/add features, write prompts, make the changes one by one, and maintain separate commits for each task.

So for your chat-history deliverable

This ChatGPT conversation is your Vibe Coding chat history.

You should save/export **this conversation**, because it contains the prompts you gave me and the coding work we did.

You don't need to find another mysterious "Vibe Coding chat." Ø=Þ-

Your final deliverables are essentially:

- ' Updated code
- ' Task 1 commit
- ' Task 2 commit
- ' Task 3 commit
- ' Task 4 commit
- ' BEFORE video

- ' AFTER video
- Chat history exported as PDF/DOC

And you're currently working on getting the videos into GitHub. After that, **the chat-history export is the last major item.**

If you want, I can:

- Add restart logic explanation to main.py
- Outline final submission steps
- Suggest a last-minute test scenario